Resolution Geometry: The Duality of Gravity

Structure, Energy Footprint, and the Exact Point at Which Newton Must Fail

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Abstract

A static gravitational field consists of two independent objects. The spatial metric h is what rods and radar at rest measure; we call it *structure*. The lapse N is what clocks, redshift and static support measure; we call it the *energy footprint*. We prove that free motion is a one-parameter pencil of these two: trajectories of asymptotic speed v are the geodesics of $j_v = (N^{-2} - 1 + v^2)\, h$. Four exact results follow.

(i) If h is flat, every orbit at every speed, light included, is *exactly* Newtonian in the potential $\Phi_{\text{kin}} = \frac{c^2}{2}(1 - N^{-2})$, however strong the field.

(ii) After inverse scattering, two-speed deflection data obey an *exactly affine* law, $v^2 n_v^2 = \psi^2(N^{-2} - 1) + \psi^2 v^2$, in isotropic radius. The slope is structure and the intercept is energy.

(iii) The scattering family at two speeds over all impact parameters is Newton-representable by *any* potential if and only if h is flat. Single-speed data never refute Newton.

(iv) Within the energy channel, the trajectory, support and clock potentials agree at first order and split at second. The support/trajectory force ratio is exactly N^2.

The failure point of Newton is therefore exact and typed. Structure breaks Newton at any field strength, as soon as two probe speeds are compared below the precision threshold $\varepsilon_{\text{crit}} = \gamma|v_1^2 - v_2^2|/(2 + \gamma(v_1^2 + v_2^2))$, which is radius-independent in the weak field. Energy breaks Newton's single potential where $1 - N^2$ exceeds the declared tolerance.

The theorem answers a prior hostile attack. That attack refuted an exact αv^2 test, showed that vanishing structural deflection does not force αv^2 scaling, and built a two-speed collision at a single impact parameter. All three are absorbed: the first two are artefacts of the nonlinear angle–index transform, and the third is a finite-panel correspondence that a functional panel resolves. The battery runs 14 attacks, all locked, and rejects 6 canaries (master seed 20261024).

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1 The thesis and what failed before

The thesis is that gravity has two forms, a structural one and an energy footprint; that Newton works where they overlap; and that Newton fails where they diverge. Earlier formalizations were not robust. A hostile battery (41 assertions, inherited here) established four points.

(a) Attributing a single static observable, the support force, to "structure" or to "energy" depends on convention. In Schwarzschild $NB = 1$, so the same factor reads either way, and areal and proper radius give different splits.

(b) The proposal $\alpha v^2 = \text{const}$ for Newton is exact only to first order. Exact Coulomb scattering $\alpha = 2\arctan(\mu/bv^2)$ violates it.

(c) With flat h and $N^2 = 1 - 2m/r$, the Gauss–Bonnet structural term vanishes, yet αv^2 is not constant.

(d) A three-parameter competitor metric matches Schwarzschild at two speeds and a single impact parameter.

This paper replaces formula-splitting with an *exact pencil identity*. It replaces the αv^2 slope with an *exact affine law after inversion*, and it replaces finite panels with *functional panels*. All four findings above become consequences of the theory rather than threats to it.

2 Declarations

Units $G = c = 1$; m denotes GM/c^2.

Hypothesis 2.1 (H-st)**.** A static, spherically symmetric, asymptotically flat exterior $ds^2 = -N^2dt^2 + h$ with $h = B^2dr^2 + r^2d\Omega^2$ in areal radius. The static Killing congruence is the declared observer family.

Hypothesis 2.2 (H-geo)**.** Free test bodies follow timelike geodesics and light follows null geodesics. This is the equivalence-principle coupling, declared rather than derived.

Hypothesis 2.3 (H-mon)**.** No orbiting: for the speeds used, $w = \rho\, n_v(\rho)$ is strictly increasing on the probed region.

Hypothesis 2.4 (H-pan)**.** A *functional* panel: the deflection $\alpha(b, v_i)$ is known for all $b > b_*$ at each declared speed $v_i \in (0, 1]$. Here v is the asymptotic speed and b the asymptotic impact parameter.

Definition 2.5 (Newtonian class)**.** Flat space δ and a central potential $\Phi(r) \to 0$. Galilean kinematics: energy $\varepsilon = v^2/2$ and angular momentum vb per unit mass. Every test body falls identically.

Definition 2.6 (The two channels)**.** *Structure* is h, written in isotropic form $h = \psi(\rho)^2(d\rho^2 + \rho^2 d\Omega^2)$, which exists for every spherical h. *Energy footprint* is N. It fixes redshift, the support acceleration $a = c^2|\nabla \ln N|_h$, and the Killing energy $E_\infty = mc^2N$.

3 The duality: free motion is a pencil

Theorem 3.1 (Jacobi pencil)**.** *Under H-st and H-geo, the spatial trajectories of free bodies with asymptotic speed v are the unparametrized geodesics of*

$$j_v = (N^{-2} - \kappa)\, h, \qquad \kappa = 1 - v^2.$$

The family $\{j_v\}$ is spanned by h (structure) and the optical metric h/N^2 ($v = 1$).

Proof. For Killing energy E and mass m, trajectories are geodesics of the Jacobi metric $(E^2/N^2 - m^2)h$ [1]. With $E = m/\sqrt{1-v^2}$ this equals $\frac{m^2}{1-v^2}(N^{-2} - \kappa)h$, and a constant factor does not change geodesics. Light is the limit $m \to 0$, i.e. $\kappa = 0$ (battery D01). □

Proposition 3.2 (Local law; signed)**.** *Let T be the unit tangent of a trajectory in h and v_{loc} the speed measured by static observers. Then*

$$\nabla^h_T T = -\frac{c^2}{v_{\mathrm{loc}}^2}\,(\mathrm{grad}_h \ln N)_\perp.$$

The path bends toward decreasing N, with Newton's centripetal form, but measured in h.

Proposition 3.3 (Gauss–Bonnet split, relative to the static congruence)**.** *On simple asymptotic scattering paths, $\alpha = A_E + A_S$ with $A_E = \int_\gamma \kappa_h \, d\ell_h$ and $A_S = -\iint_D K_h \, dA_h = \int_\gamma (1 - B^{-1})\, d\phi$. Moreover $A_S \equiv 0$ when h is flat, and at first order $A_S/A_E = \gamma v^2$ (battery D11, residual 8×10^{-12}). This is a per-path diagnostic; the exact global criterion is Theorem 4.4.*

4 Exact Newtonian representability

Theorem 4.1 (Flat structure is exactly Newtonian)**.** *If h is flat, the trajectory family at* every *speed $v \in (0, 1]$ coincides with the Newtonian family in*

$$\Phi_{\mathrm{kin}} = \tfrac{c^2}{2}\big(1 - N^{-2}\big),$$

with Galilean kinematics, at any field strength. Light ($v = c$) is Newton's corpuscle at speed c.

Proof. For $h = \delta$, $N^{-2} - \kappa = v^2 + (N^{-2} - 1) = 2(\frac{v^2}{2} - \Phi_{\mathrm{kin}})$. This is Newton's Jacobi metric $2(\varepsilon - \Phi)\delta$ with $\varepsilon = v^2/2$. □

The battery (D03) integrates the full four-dimensional geodesic equations and Newton's equations independently, for $N^2 = 1 - 2m/r$ and for $N = e^{-m/r}$. The exit angles agree to 1.9×10^{-9} rad for deflections up to 1.44 rad. Hence the hostile flat-spatial canary (c) does not show non-Newtonian motion. It shows that $\Phi_{\mathrm{kin}} = -m/(r - 2m)$ is not a Coulomb potential (D08).

Theorem 4.2 (Exact affine law)**.** *In isotropic radius, the refractive index n_v of j_v, normalized to* 1 *at infinity, satisfies*

$$\boxed{v^2\, n_v(\rho)^2 = \psi(\rho)^2\big(N(\rho)^{-2} - 1\big) + \psi(\rho)^2\, v^2}$$

exactly, for every v. For a Newtonian potential, $v^2 n_v^2 = v^2 - 2\Phi$, so the slope is 1.

Proof. $j_v = \psi^2(N^{-2} - 1 + v^2)(d\rho^2 + \rho^2 d\phi^2)$ in the equatorial plane; divide by the asymptotic value v^2. □

Imported Theorem 4.3 (Fixed-energy classical inverse scattering [3, 4])**.** *Under H-mon, the deflection function at one speed determines the index profile uniquely:*

$$\ln n(s) = \frac{1}{\pi}\int_0^\infty \alpha(s\cosh t)\,dt, \qquad \rho = s/n(s).$$

Theorem 4.4 (Newton representability $\Leftrightarrow$ flat structure)**.** *Assume H-st, H-geo, H-mon, and H-pan at two distinct speeds $v_1 \neq v_2$. Then some Newtonian potential reproduces $\alpha(\cdot, v_1)$ and $\alpha(\cdot, v_2)$ on $b > b_*$ if and only if $\psi \equiv 1$ on the probed region, i.e. h is flat there. In that case $\Phi = \Phi_{\mathrm{kin}}$, uniquely.*

Proof. ($\Leftarrow$) is Theorem 4.1. ($\Rightarrow$) By Imported Theorem 4.3, equal deflection functions at v_i force equal index profiles as functions of the flat-chart radius. By Theorem 4.2, $\psi^2(N^{-2}-1) + \psi^2 v_i^2 = v_i^2 - 2\Phi$ for $i = 1, 2$. Subtracting gives $(\psi^2 - 1)(v_1^2 - v_2^2) = 0$, so $\psi^2 = 1$, and then $\Phi = \frac{1}{2}(1 - N^{-2})$. □

Corollary 4.5 (Single-speed no-go)**.** *For every static spacetime and every single speed v, the potential $\Phi_v = \frac{v^2}{2}(1 - n_v^2)$ reproduces $\alpha(\cdot, v)$ exactly. No single-speed scattering experiment can refute Newton. Battery D10 reproduces Schwarzschild at $v = 0.5$ to 5.8×10^{-7}.*

Corollary 4.6 (Exact two-speed identification of both channels)**.** *From two functional panels,*

$$\psi^2 = \frac{v_1^2 n_1^2 - v_2^2 n_2^2}{v_1^2 - v_2^2}, \qquad N^{-2} = 1 + \frac{v_1^2 n_1^2 - \psi^2 v_1^2}{\psi^2},$$

and every further speed is predicted exactly.

The battery (D05, D06) reconstructs structure and energy for six metrics: Schwarzschild, $\gamma = -0.5, 0.4, 1.6$, a nonlinear (q, β) case, and an exponential lapse. Errors relative to the signal are at most 1.2×10^{-7} for structure and 5.6×10^{-8} for energy. Held-out speeds 0.3, 0.7 and light are predicted to 3.3×10^{-8} of $n - 1$. Mutating the kinematic weight from v^2 to v breaks the prediction by 8.4×10^{-3} (D14).

Remark 4.7 (Why the earlier tests were weak)*.* The remainder $R(b, v)$ that contaminated the αv^2 slope belongs to the nonlinear map $\alpha \leftrightarrow n$, not to the law. After inversion the law is exactly affine. Exact Coulomb data return $|\psi^2 - 1| \leq 10^{-9}$, although their αv^2 slope is 2.7×10^{-4} (D07). The hostile single-b collision is reproduced to 1.3×10^{-13}. The two metrics then differ at $b = 30, 100, 1000$ by 2.8×10^{-4}, 1.1×10^{-4} and 1.5×10^{-5}, and full-b reconstruction separates their structures by 1.6×10^{-4} against a floor of 3×10^{-9} (D09). In RG terms, a finite panel is a CORRESPONDENCE and a functional panel is a LOCK.

Proposition 4.8 (Newton's miss is the structure)**.** *Take the single-speed potential fitted at v_1 and use it to predict the deflection at v_2. Its index error is exactly*

$$v_2^2\big(n_{v_2}^2 - n_{\mathrm{N}}^2\big) = (\psi^2 - 1)(v_2^2 - v_1^2).$$

In the weak field, the angle miss equals $A_S(v_2)\,(1 - v_1^2/v_2^2)$. The battery (D10) matches this to 3.2%, 1.0% and 0.1% at $b = 60, 200, 2000$.

Proof. Write $P = \psi^2(N^{-2} - 1)$. Then $n_{\mathrm{N}}^2 = 1 + (v_1^2/v_2^2)(n_1^2 - 1)$ and $v_1^2 n_1^2 = P + \psi^2 v_1^2$, so $v_2^2 n_{\mathrm{N}}^2 = v_2^2 + P + \psi^2 v_1^2 - v_1^2$. Subtract this from $v_2^2 n_{v_2}^2 = P + \psi^2 v_2^2$. □

5 The energy channel: overlap and divergence

Theorem 5.1 (Three energy potentials)**.** *The trajectory, support and clock potentials,*

$$\Phi_{\text{kin}} = \tfrac{c^2}{2}(1 - N^{-2}), \qquad \Phi_{\text{hold}} = c^2 \ln N, \qquad \Phi_{\text{clock}} = c^2(N-1),$$

satisfy, with $x = 1 - N$*,*

$$\Phi_{\text{kin}}/c^2 = -x - \tfrac{3}{2}x^2 + \dots, \qquad \Phi_{\text{hold}}/c^2 = -x - \tfrac{1}{2}x^2 + \dots, \qquad \Phi_{\text{clock}}/c^2 = -x.$$

They overlap at first order and diverge at second. On flat h*, the ratio of support force to trajectory force is exactly* $d\Phi_{\text{hold}}/d\Phi_{\text{kin}} = N^2$*. A single Newtonian potential reproduces both orbits and static support if and only if* N *is constant (D02, D13).*

Example 5.2. $N = e^{-m/r}$ has support acceleration exactly m/r^2, which is Newtonian, and exactly Newtonian orbits in Φ_{kin}. The two potentials still disagree by the factor N^2: 0.513 at $r = 3m$ and 0.9998 at $r = 10^4 m$.

This is the precise content of "they overlap in weak fields and diverge in strong fields". It is exactly true, but only for the *energy* channel.

6 The exact failure point

Theorem 6.1 (Duality failure theorem)**.** *Let a panel declare relative deflection errors* $|\delta\alpha| \le \varepsilon\,\alpha$ *at two speeds, and/or support and orbit measurements with fractional tolerance* δ*.*

(I) ***Structural failure.*** *The inversion is linear and positive, so* $|\delta \ln n_i| \le \varepsilon \ln n_i/(1-\varepsilon)$*. Hence*

$$|\hat{\psi}^2 - \psi^2| \le \beta(\rho) = \frac{\sum_i v_i^2 n_i^2 \operatorname{expm1}\big(2\varepsilon \ln n_i/(1-\varepsilon)\big)}{|v_1^2 - v_2^2|}.$$

Newton is refuted at ρ *(status INCONSISTENT) iff* $|\hat{\psi}^2 - 1| > \beta(\rho)$*. In the weak field,* $\psi^2 - 1 \approx 2\gamma m/\rho$ *and* $\beta \approx 2\varepsilon m(2 + \gamma(v_1^2 + v_2^2))/(\rho|v_1^2 - v_2^2|)$*. Both scale as* $1/\rho$*, so*

$$\boxed{\textit{Newton fails structurally} \iff \varepsilon < \varepsilon_{\text{crit}} = \frac{\gamma\,|v_1^2 - v_2^2|}{2 + \gamma(v_1^2 + v_2^2)}}$$

at every radius, however weak the field (stated for $\gamma > 0$*).*

(II) ***Energetic failure.*** *Newton's single potential fails for orbits plus support where* $1 - N^2 > \delta$*; for* $N^2 = 1 - 2m/r$ *this is* $r < 2m/\delta$*.*

Proof. (I) The Firsov map is linear with a positive kernel; $\alpha > 0$ under H-mon. Propagate through Corollary 4.6, and expand n_i using Theorem 4.2 with $\psi^2 \simeq 1 + 2\gamma m/\rho$ and $N^{-2} \simeq 1 + 2m/\rho$. (II) is Theorem 5.1. □

Battery D12 uses Schwarzschild data at $v = 0.5$ and light, for which $\varepsilon_{\text{crit}} = 0.2308$, with seeded smooth noise. At $0.5\,\varepsilon_{\text{crit}}$ Newton is refuted at every radius from $40m$ to $10^4 m$; at $2\,\varepsilon_{\text{crit}}$ it is refuted at none. The bound covered the true structure in both runs.

For a very slow probe compared with light, $\varepsilon_{\text{crit}} \to \gamma/(2+\gamma) = \frac{1}{3}$ in general relativity. A joint relative precision better than about 30%, on light deflection and on slow-orbit calibration of the source, is enough to resolve structure. This is the logic of the 1919 test, in which the general-relativity value is twice Soldner's Newtonian one [7]; we do not reanalyse those data.

Remark 6.2 (The thesis, restated as a theorem). Newton works where the panel cannot resolve either channel. It fails exactly when one becomes resolved.

- *Structure* breaks Newton at any field strength. It is invisible to single-speed panels (Corollary 4.5) and to slow probes, where its share of the deflection is γv^2. It becomes visible once two speeds are compared below $\varepsilon_{\text{crit}}$.
- *Energy* never breaks Newton's orbits (Theorem 4.1). It breaks the single-potential bookkeeping between orbits, support and clocks at fractional size $1 - N^2$, which is weak-field overlap and strong-field divergence.

The parameter γ, the ratio of structure to energy, is *measured*, not derived. The Cassini value is $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ [6].

7 Placement in RG

Probe speed is a protocol parameter. A single-speed functional panel leaves structure and energy in one fibre (CORRESPONDENCE). Two speeds form the minimal sufficient panel for (h, N) (LOCK), and finite panels at fixed b are again correspondences. The construction is theory-neutral: it uses only (h, N) and H-geo. It neither derives Einstein's equation nor contradicts the source-remainder no-go of the physical path paper [8]. It types Newton's failure.

Rung	Statement	Status	Witness
$\Psi 0$	Declarations: H-st, H-geo, H-mon, H-pan; Newtonian class	declared	—
$\Psi 1$	Jacobi pencil $j_v = (N^{-2} - 1 + v^2)h$; duality (h, N)	F	D01
$\Psi 2$	Signed local law $\nabla_T T = -(c^2/v^2)(\nabla \ln N)_\perp$	F	hostile
$\Psi 3$	Gauss–Bonnet split, relative to the static congruence	F	D11
$\Psi 4$	Newton scaling: orbit shape depends on Φ/v^2; αv^2 constant only at first order	F / N	D07
$\Psi 5$	Flat $h \Rightarrow$ exactly Newtonian in Φ_{kin}, any strength	F	D03, D08
$\Psi 6$	Exact affine law $v^2 n_v^2 = P + \psi^2 v^2$	F	D04–D06
$\Psi 7$	Newton-representable $\Leftrightarrow$ h flat (two functional speeds)	F (I: Firsov)	D05
$\Psi 8$	Single-speed no-go	N	D10
$\Psi 9$	Exact two-speed identification; held-out prediction	F	D05, D06, D14
$\Psi 10$	Finite panels collide (correspondence); functional panels lock	N / F	D09
$\Psi 11$	Newton's miss $= (\psi^2 - 1)(v_2^2 - v_1^2)$	F	D10
$\Psi 12$	Energy channel: three potentials; support/orbit ratio N^2	F	D02, D13
$\Psi 13$	Failure point: $\varepsilon_{\text{crit}}$ (structure), $1 - N^2 > \delta$ (energy)	F	D12, D13
$\Psi 14$	γ is measured, not derived	declared / data	[6]
$\Psi 15$	Stationary (frame dragging), time-dependent, non-spherical fields; real data	O	—

8 Battery

The scripts `rg_duality_attacks.py` and `rg_duality_core.py` (NumPy, SciPy, SymPy; master seed 20261024) run 14 attacks, all locked, and reject 6 canaries. A manifest with the source hash is written before evaluation, and two fresh-process runs give identical witnesses.

The six canaries are false statements, all rejected:

(1) a nonzero αv^2 slope proves non-Newtonian gravity;

(2) single-speed data can refute Newton;

(3) $A_S = 0$ forces αv^2 constancy;

(4) two speeds at one impact parameter identify the geometry;

(5) one potential serves orbits and support on flat h;

(6) the affine law holds with kinematic weight v.

Exact arithmetic is used for the pencil and the potential series. Deflections come from cancellation-free radial quadrature and are cross-checked by independent four-dimensional geodesic and Newton ODEs. The inverse scattering is validated against exact Coulomb data to 3×10^{-12}. The prior hostile battery (41/41) is included for provenance.

Limits. The battery is not blind. All data are synthetic. The functional panel is approximated by 700 impact parameters with an analytic tail, and the noise model is smooth and bounded.

9 Non-claims

1. No derivation of general relativity or of γ; no empirical test performed.
2. Scope: static, spherical, asymptotically flat, no orbiting, test bodies, geodesic coupling. Stationary fields add a third channel (gravitomagnetism) that is not treated here.
3. The split into structure and energy is relative to the static congruence, which is canonical for static spacetimes but is a declared observer family.
4. Theorem 4.4 needs a functional panel. Finite panels only bound a correspondence.
5. The classical inverse-scattering and Jacobi-metric results are imported; novelty is claimed only for their assembly into the duality and failure theorems.

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